

Stage 1: Multi-Source Literature Retrieval and Deduplication I

We retrieve papers from three complementary academic databases to maximize coverage and enable cross-source deduplication. The retrieval window begins at 2005, encompassing the period when Energy-Based Model and variational Bayesian research [dayan1995helmholtz, lecun2006tutorial] provided mathematical precursors to what Friston formalized as the Free Energy Principle in 2006 [friston2006free]; this inclusive start captures historical lineage and cross-disciplinary convergence that a later cutoff would exclude.

Stage 1: Multi-Source Literature Retrieval and Deduplication II

arXiv. We query the arXiv Atom API using five phrase-matched searches by default: `all:"active inference"`, `all:"free energy principle"`, `all:"predictive coding" AND all:"free energy"`, `all:"expected free energy"`, and `all:"variational free energy" AND all:"inference"`. The `all:` prefix searches titles, abstracts, and full text; phrase matching reduces contamination from unrelated physics papers that mention “free energy” in thermodynamic contexts. Additional Energy-Based Model queries (`all:"energy-based model" AND all:"free energy"`, `all:"Helmholtz machine" AND all:"inference"`, `all:"Boltzmann machine" AND all:"free energy"`, `all:"contrastive divergence" AND all:"generative model"`) are available via the `arxiv_queries` list in `config.yaml` for researchers wishing to capture adjacent EBM literature at the intersection of energy-based generative modeling and variational inference [lecun2006tutorial].

Stage 1: Multi-Source Literature Retrieval and Deduplication III

Semantic Scholar. We query the Semantic Scholar Graph API [kinney2023semantic] with the same terms. Semantic Scholar provides citation graphs, abstract embeddings, and links to published versions. Retry logic with exponential backoff handles rate limiting.

OpenAlex. We query OpenAlex [priem2022openalex] to capture journal-published work that may not appear on arXiv, including clinical studies and neuroscience experiments in domain-specific venues. The `referenced_works` field populates citation links for each paper.

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Canonical Identifier Deduplication

After retrieval, papers are assigned a canonical identifier using the priority scheme: DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID > title hash. When the same paper appears in multiple sources, the record with the highest metadata completeness is retained. For each incoming paper, the two records are compared on metadata completeness—defined as the count of non-empty optional attributes across the full Paper record (abstract, DOI, arXiv ID, Semantic Scholar ID, OpenAlex ID, venue, citation count, references, publication date, PDF URL, open-access flag, and author list). The pipeline retains the richer record; in the event of a tie, the incumbent is preserved. This “merge-on-add” strategy aggregates the richest available metadata without requiring an expensive downstream reconciliation pass. Deduplication produces $N = 817$ unique papers spanning 2005–2026.

Stage 1: Multi-Source Literature Retrieval and Deduplication V

Relevance Filtering and Curation

After deduplication, a **relevance filter** removes papers whose titles and abstracts lack any core Active Inference terminology (e.g., active inference, 'free energy principle,' 'variational free energy'), eliminating off-topic results introduced by broad keyword overlap across heterogeneous databases. We acknowledge that this retrieval strategy yields limited bibliographic depth, functioning as a representative snapshot rather than an exhaustive census of the literature.

We emphasize that this process relies on keyword search strategies across divergent APIs. In any complex research field, there is no single optimal word or threshold for definitive inclusion or exclusion. Different information sources and repositories yield differing schemas and representations, introducing both false positives (e.g., machine learning papers that mention “free energy” in a purely thermodynamic context or bioinformatics tools whose names